

Practice Exam 4 (A)

(See also the fluid pressure problems on the Koen practice exam.)

1 What is the simplest shape that represents the total loading distribution on plate AB?

- A. trapezoid
- ☒ B. triangle
- C. square
- D. rectangle
- E. parabolic curve

2 Where would the center of pressure for the hydrostatic force acting on gate AB be located along AB measured from A?

$h = 2 \text{ m}$

- A. 1.67 m
- B. 0.83 m
- C. 1.12 m
- D. 2.24 m
- E. 0.75 m

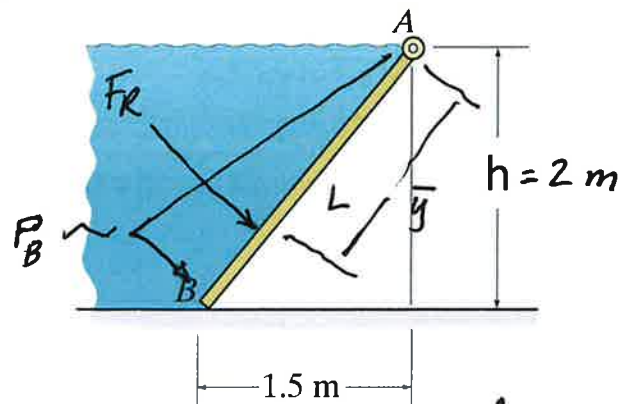
If $\gamma = 9810 \text{ N/m}^3$

and gate is 1 m deep,

$$P_B = 2\gamma = 19.62 \text{ kN/m}^2$$

$$F_R = \frac{1}{2}(19.62)(2.5)(1 \text{ m wide})$$

$$F = 24.5 \text{ kN}$$



$$L = 2.5 \text{ m}$$



$$\bar{y} = \frac{2}{3}(2.5) = 1.67 \text{ m from A}$$

Section 2a Moment of Inertia

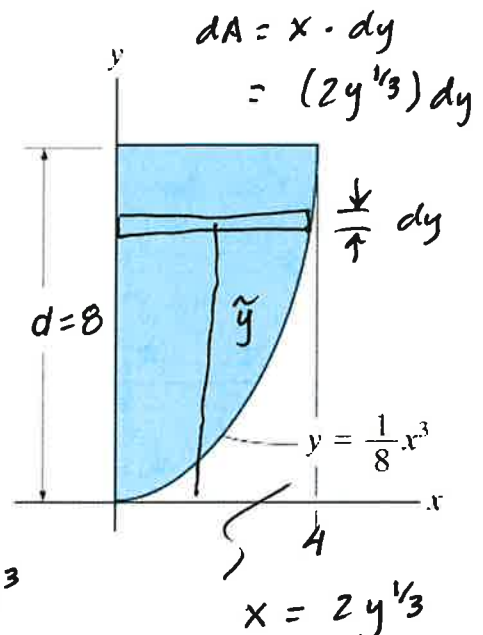
3. Determine the moment of inertia of the shaded area about the x axis. $d = 8 \text{ m}$

- A. 538 m⁴
- ☒ B. 614 m⁴
- C. 702 m⁴
- D. 910 m⁴
- E. 869 m⁴

$$I_x = \int_A \tilde{y}^2 dA$$

$$= \int_0^8 y^2 (2y^{1/3}) dy$$

$$= \frac{3}{10} 2 y^{10/3} \Big|_0^8 = 614.4 \text{ m}^4 = I_x$$



4. Which initial expression would be integrated to determine the moment of inertia with respect to the y axis when using a horizontal differential element and single integration for the shaded area?

- A. ($x^2 dy$)
- B. ($y^2 dx$)
- C. ($.125 x^3 dx$)
- ☒ D. ($x^3 / 12 + x^3 / 4$) dy
- E. ($x^3 / 12 - x^3 / 4$) dy

$$I_y = \int_A dI_y = \int_A \frac{1}{3} b h^3$$

$$= \int_0^8 \frac{1}{3} (2y^{1/3})^3 dy$$

$$b = dy$$

$$h = x = 2y^{1/3}$$

$$= \int_0^8 \frac{8}{3} y dy = \int_0^8 \frac{8}{3} \left(\frac{1}{8} x^3 \right) dy = \int_0^8 \frac{1}{3} x^3 dy$$

$$= \int_0^8 \left(\frac{x^3}{12} + \frac{x^3}{4} \right) dy$$

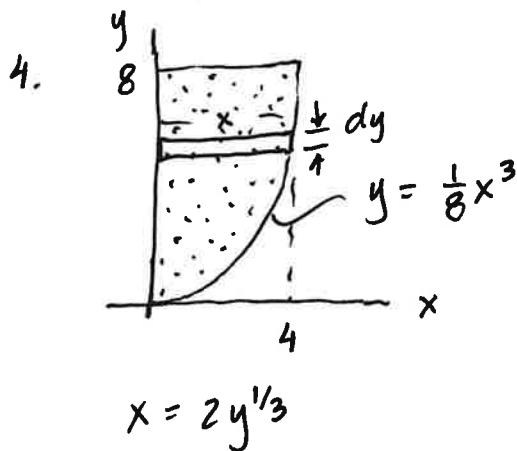
Crazy Problem!

$$\frac{8}{6} y^2 \Big|_0^8 = \frac{8 \cdot 64}{6} = 85.3 \text{ m}^4 = I_y$$

In terms of x and dy?!

$$\frac{1}{12} + \frac{1}{4} = \frac{1}{12} + \frac{3}{12} = \frac{4}{12} = \frac{1}{3}$$

More on Problem 4



Find the initial expression for I_y , when using a horizontal element.

I used $dI_y = \frac{1}{3}bh^3$ which I think is the best approach for this element.

But evidently they used the full PAT:

$$\begin{aligned} dI_y &= d\bar{I} + dA \cdot \tilde{x}^2 \\ &= \frac{1}{12}(x^3)dy + x\left(\frac{x}{2}\right)^2 dy \\ &= \left[\left(\frac{1}{12}x^3\right) + \frac{1}{4}x^3\right] dy \end{aligned}$$

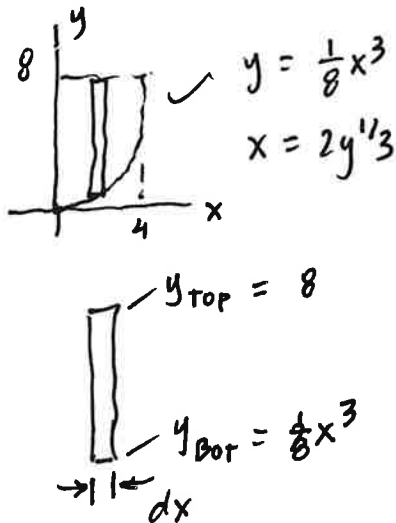
$$\begin{aligned} d\bar{I} &= \frac{1}{12}bh^3 \\ &= \frac{1}{12}(x^3)dy \end{aligned}$$

$$dA = x \cdot dy \quad \tilde{x} = \frac{x}{2}$$

Thus, $I_y = \int_0^8 dI_y$

but still need to plug $x = 2y^{1/3}$ in for $x \dots$

If used element || to y axis:



$$\begin{aligned} I_y &= \int_A \tilde{x}^2 dA = \int_0^4 x^2 \left(8 - \frac{1}{8}x^3\right) dx \\ &= \int_0^4 \left(8x^2 - \frac{1}{8}x^5\right) dx \\ &= \left[\frac{8}{3}x^3 - \frac{1}{48}x^6\right]_0^4 \end{aligned}$$

$$I_y = 85.33 \text{ m}^4$$

The composite shape can be viewed as a rectangle plus a triangle, with a circle subtracted out.

The parallel axis theorem ($I = \bar{I} + Ad^2$) can be used to determine the moment of inertia about the x axis.

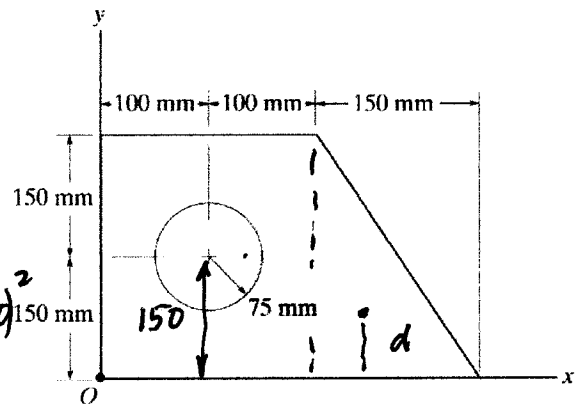
5. What is the magnitude of I_x for the circle? *No sign needed.*

- A. $3.10 (10^8) \text{ mm}^4$
- B. $2.02 (10^8) \text{ mm}^4$
- C. $5.31 (10^8) \text{ mm}^4$
- D. $1.73 (10^8) \text{ mm}^4$
- ✓ E. $4.22 (10^8) \text{ mm}^4$

$$I_x = \bar{I} + Ad^2$$

$$= \frac{\pi}{4}(75)^4 + \pi(75)^2(150)^2$$

$$= 422.5 \cdot 10^6 \text{ mm}^4$$



6. What is $\bar{I}$ for the rectangle? (about centroidal axis $\parallel$ to x...)

- ✓ A. $450 (10^6) \text{ mm}^4$
- B. $200 (10^6) \text{ mm}^4$
- C. $300 (10^6) \text{ mm}^4$
- D. $350 (10^6) \text{ mm}^4$
- E. $275 (10^6) \text{ mm}^4$

$$\bar{I} = \frac{1}{12}(200)(300)^3$$

$$\bar{I} = 450 \cdot 10^6 \text{ mm}^4$$

7. What is "d" for the triangular area? (If use I_{base} for Δ , do not need d)

- A. 50 mm
- B. 200 mm
- C. 250 mm
- ✓ D. 100 mm
- E. 300 mm

If use PAT, $I_x = \bar{I} + Ad^2$

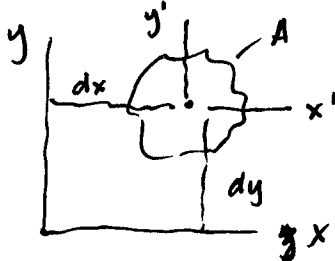
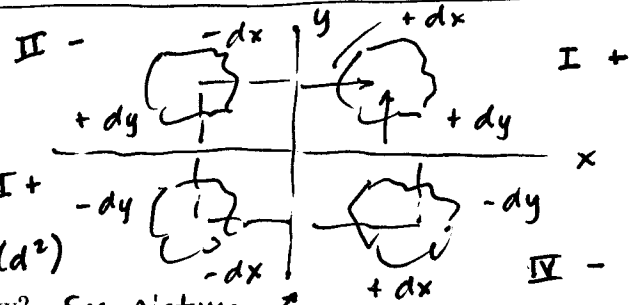
$$d = \frac{300}{3} = 100 \text{ mm} = \frac{1}{36}bh^3 + \left(\frac{1}{2}bh\right)(d)^2$$

$$I_{\text{base}} = \frac{1}{12}bh^3$$

Section 3 – Product of Inertia

POI Concepts:

1. What are units of MOI? POI? $L^4 \begin{bmatrix} \text{in}^4 \\ \text{cm}^4 \\ \text{mm}^4 \end{bmatrix}$
2. How do areas (A) and distance (d) affect MOI? POI? $f(d^2)$
3. Does the location with respect to the axes affect POI? How? See picture
4. If the area is more in quadrant I or II or III or IV—how will each affect POI?
5. What if an area is symmetric, how does that affect POI? If symmetric about either x or y then $I_{xy} = 0$.
6. What do values of POI mean? What if POI for a shape (and for a set of axes) is zero?
7. Understand each term of the POI PAT: $I_{xy} = \bar{I}_{x'y'} + A dx dy$
8. What are possible values of $\bar{I}_{x'y'}$? Of dx and/or dy?

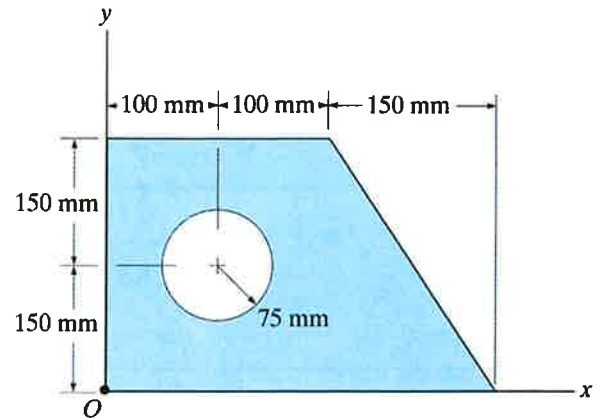


$\bar{I}_{x'y'} = 0$
for symmetric shapes.
Not zero for Δ 's or D's...

If $I_{xy} = 0$, then I_x, I_y are principals. (MAX, MIN)

★ But, there are some areas which are NOT symmetric which have $I_{xy} = 0$. Special cases.

9. For the shape at right, what are $\bar{I}_{x'y'}$ and A and dx and dy values for each of the component shapes?



Shape	$\bar{I}_{x'y'}$	A	dx	dy
Rectangle	0	$200(300)$	100	150
Circle	0	$\pi(75)^2$	100	150
Triangle	$-\frac{1}{72}b^2h^3$	$\frac{1}{2}(150)(300)$	$200 + 50$	100

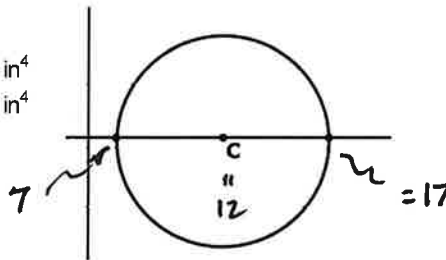
$$= -\frac{1}{72}(150)^2(300)^2$$

Mohr's Circle

Question 1

Given I_{AVG} and diameter, find I_{MAX} and I_{MIN} .

Center I_{AVG} =	12.00	in ⁴
Diameter d =	10.00	in ⁴



$$R = \frac{10}{2} = 5 \text{ in}^4$$

$$I_{MAX} = 12 + 5 = 17 \text{ in}^4$$

$$I_{MIN} = 12 - 5 = 7 \text{ in}^4$$

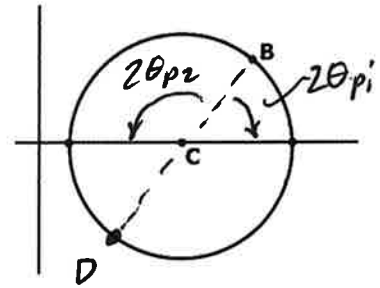
Question 2

Given I_{AVG} and point B $[I_u, I_{uv}]$:

Center I_{AVG} =	12.00	in^4
Coordinates of point B $[I_B, I_{BD}]$:	I_B =	19.88 in^4
	I_{BD} =	6.16 in^4

Please determine:

Radius R , I_{MAX} , I_{MIN} , point D $[I_D, I_{BD}]$ (90 deg from point B on area), θ_{p1} , θ_{p2}



$$R = 10 \text{ in}^4$$

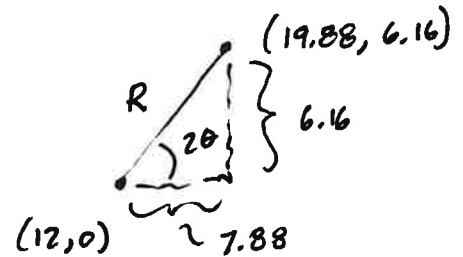
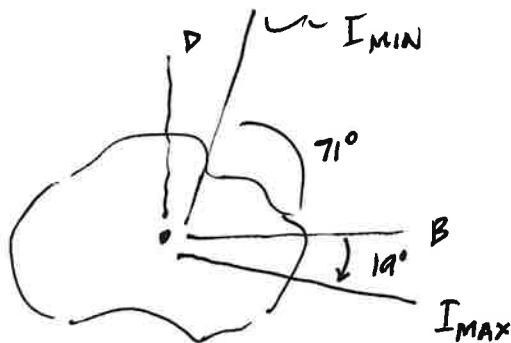
$$I_{MAX} = 22 \text{ in}^4$$

$$I_{MIN} = 2 \text{ in}^4$$

$$I_D = 12 - 7.88$$

$$= 4.12 \text{ in}^4$$

$$D: [4.12, -6.16]$$



$$R = \sqrt{7.88^2 + 6.16^2}$$

$$R = 10$$

$$2\theta_{p1} = \tan^{-1} \frac{6.16}{7.88}$$

$$= 38$$

$$\theta_{p1} = -19^\circ$$

$$2\theta_{p2} = 180 - 38$$

$$= 142^\circ$$

$$\theta_{p2} = 71^\circ$$